

Name: \_\_\_\_\_

# PLACE VALUE AND ORDERING INTEGERS, (ESTIMATING CALCULATIONS)

Date: \_\_\_\_\_

**LO: I can estimate the answer to calculations by rounding to appropriate values.**

## Estimation

**Estimating** means finding an **approximate answer** by rounding numbers to 1 significant figure (or to convenient values) before calculating.

**Purpose:** Estimation helps check calculator answers and gives a quick sense of the size of an answer.

## Quick examples

|   |                         |                       |                            |
|---|-------------------------|-----------------------|----------------------------|
| ● | $4.8 \times 21.3$       | $5 \times 20 = 100$   | round to 1 sf              |
| ● | $198 \div 4.1$          | $200 \div 4 = 50$     | round to convenient        |
| ● | 26                      | 5 (since $5^2 = 25$ ) | nearest square             |
| ● | $4.8 \times 21.3$       | $4 \times 21 = 84$    | round to 1 sf not truncate |
| ● | Estimates must be exact |                       | they are approximate       |

## Key idea

Round each number to 1 significant figure, then calculate.  
Use to show approximation.

$$4.8 \times 21.3$$

$$5 \times 20 = 100$$

**Round first, then do the simple calculation**

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - multiplication

Estimate  $6.2 \times 48$ .

Round: 6.2   6, 48   50

$$6 \times 50$$

$$= 300$$

**300**

Round to 1 sf then multiply.

### Example 2 - division

Estimate  $312 \div 5.8$ .

Round: 312   300, 5.8   6

$$300 \div 6$$

$$= 50$$

**50**

Round to 1 sf then divide.

### Example 3 - combined

Estimate  $(4.9 \times 11.2) \div 2.3$ .

Round:  $5 \times 11 \div 2$  or  $5 \times 10 \div 2$

$$= 50 \div 2$$

$$= 25$$

**25**

Round all values, then apply BIDMAS.

### Example 4 - square roots

Estimate  $\sqrt{85}$ .

$$81 = 9^2, 100 = 10^2$$

85 is between 81 and 100, closer to

81

9.2

**9.2**

Find the nearest perfect squares above and below.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Estimating multiplications

Estimate each by rounding to 1 sf.

- a)  $3.8 \times 52$
- b)  $7.1 \times 29$
- c)  $19.6 \times 4.8$
- d)  $48 \times 0.52$

### 2. Estimating divisions

Estimate each.

- a)  $198 \div 4.1$
- b)  $623 \div 19.7$
- c)  $89.4 \div 3.2$
- d)  $4750 \div 48$

### 3. Combined operations

Estimate each.

- a)  $(5.2 \times 38) \div 9.7$
- b)  $(82 + 19) \div 4.8$
- c)  $3.9^2 \times 5.1$
- d)  $(6.1 \times 8.9) \div (2.1 \times 3.8)$

### 4. Square roots

Estimate each square root.

- a) 50
- b) 30
- c) 120
- d) 200

### 5. Checking answers

Estimate to check if the given answer is reasonable.

- a)  $4.8 \times 51 = 244.8$ . Reasonable?    b)  $312 \div 5.8 = 5.38$ . Reasonable?    c)  $67 = 8.19$ . Reasonable?    d)  $19.3 \times 4.7 = 90.71$ . Reasonable?    e)  $495 \div 0.51 = 97.1$ . Reasonable?

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Estimation uses rounding to 1 significant figure ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b) Estimates should be exact ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) 50 is between 7 and 8 ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d) 4.8 rounds to 4 (1 sf) ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) Estimation helps check calculator answers ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A hall seats **48 rows of 32 chairs**. Tickets cost **£19.50** each. Estimate the maximum revenue if the hall is full.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) Estimate  $(3.7^2 \times 99) \div 1.97$ . Then calculate the exact answer and find the percentage error of your estimate.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

